Ellipses

ANSWER KEY

Section 1: Horizontal and Vertical Standard Forms

1 Find the center, vertices, and co-vertices of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. **Center:** $(0, 0)$**, Vertices:** $(\pm 5, 0)$**, Co-vertices:** $(0, \pm 3)$ ✓

b. Center: $(0, 0)$, Vertices: $(0, \pm 5)$, Co-vertices: $(\pm 3, 0)$

c. Center: $(5, 3)$, Vertices: $(\pm 5, 0)$

d. Center: $(0, 0)$, Vertices: $(\pm 25, 0)$

$a^2 = 25 \Rightarrow a = 5$ (major axis horizontal). $b^2 = 9 \Rightarrow b = 3$. Vertices at $(\pm 5, 0)$, co-vertices at $(0, \pm 3)$.

2 Find the standard equation of the ellipse with center $(0, 0)$, vertices $(0, \pm 6)$, and co-vertices $(\pm 4, 0)$.

a. $\frac{x^2}{16} + \frac{y^2}{36} = 1$ ✓

b. $\frac{x^2}{36} + \frac{y^2}{16} = 1$

c. $\frac{x^2}{4} + \frac{y^2}{6} = 1$

d. $\frac{x^2}{20} + \frac{y^2}{36} = 1$

Major axis vertical: $a = 6 \Rightarrow a^2 = 36$ under y^2. Minor axis horizontal: $b = 4 \Rightarrow b^2 = 16$ under x^2.

3 Find the length of the major and minor axes for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{4} = 1$.

a. **Major axis** $= 8$**, Minor axis** $= 4$ ✓

b. Major axis $= 4$, Minor axis $= 2$

c. Major axis $= 16$, Minor axis $= 4$

d. Major axis $= 8$, Minor axis $= 2$

$a^2 = 16 \Rightarrow a = 4$, major axis length $2a = 8$. $b^2 = 4 \Rightarrow b = 2$, minor axis length $2b = 4$.

Section 2: Foci and the Fundamental Relationship

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4 What is the fundamental relationship between semi-major axis a, semi-minor axis b, and focal distance c in an ellipse?

a. $c^2 = a^2 - b^2$ ✓

b. $c^2 = a^2 + b^2$

c. $a^2 = b^2 - c^2$

d. $c = a - b$

In an ellipse, the foci lie on the major axis inside the vertices, governed by the Pythagorean relation $c^2 = a^2 - b^2$.

5 Find the coordinates of the foci of $\frac{x^2}{25} + \frac{y^2}{9} = 1$.

a. $(\pm 4, 0)$ ✓

b. $(0, \pm 4)$

c. $(\pm\sqrt{34}, 0)$

d. $(\pm 16, 0)$

$c^2 = a^2 - b^2 = 25 - 9 = 16 \Rightarrow c = 4$. Major axis is horizontal, so foci are $(\pm 4, 0)$.

6 Find the foci of $\frac{x^2}{16} + \frac{y^2}{25} = 1$.

a. $(0, \pm 3)$ ✓

b. $(\pm 3, 0)$

c. $(0, \pm\sqrt{41})$

d. $(\pm 9, 0)$

Major axis is vertical ($a^2 = 25$). $c^2 = 25 - 16 = 9 \Rightarrow c = 3$. Foci are $(0, \pm 3)$.

Section 3: Eccentricity

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7 How is the eccentricity e of an ellipse defined, and what is its range?

a. $e = \frac{c}{a}$, **where** $0 \le e < 1$ ✓

b. $e = \frac{a}{c}$, where $e > 1$

c. $e = \frac{b}{a}$, where $e < 0$

d. $e = c^2/a^2$

Eccentricity is the ratio of focal distance to semi-major axis: $e = c/a$. For any ellipse, $0 \le e < 1$ (with $e = 0$ being a circle).

8 Find the eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{64} = 1$.

a. 0.6 ✓

b. 0.8

c. 0.36

d. 0.64

$a = 10, b = 8 \Rightarrow c = \sqrt{100 - 64} = \sqrt{36} = 6$. Eccentricity $e = c/a = 6/10 = 0.6$.

9 What happens to the shape of an ellipse as its eccentricity approaches 0?

a. **It becomes nearly circular** ✓

b. It becomes very elongated / needle-like

c. It approaches a parabola

d. It opens outward

When $e \to 0$, $c \to 0$, meaning the two foci merge into the center, producing a circle ($a = b$).

Section 4: Completing the Square and Whispering Galleries

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10 Convert $4x^2 + 9y^2 - 16x + 54y + 61 = 0$ to standard form.

a. $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{4} = 1$ ✓

b. $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{9} = 1$

c. $\frac{(x+2)^2}{9} + \frac{(y-3)^2}{4} = 1$

d. $(x-2)^2 + (y+3)^2 = 36$

$4(x^2 - 4x + 4) + 9(y^2 + 6y + 9) = -61 + 16 + 81 = 36$. Dividing by 36: $\frac{(x-2)^2}{9} + \frac{(y+3)^2}{4} = 1$.

11 A whispering gallery is an elliptical room 100 ft long and 60 ft wide. How far from the center are the two whispering spots (foci)?

a. 40 ft ✓

b. 50 ft

c. 30 ft

d. 20 ft

$2a = 100 \Rightarrow a = 50$. $2b = 60 \Rightarrow b = 30$. $c = \sqrt{50^2 - 30^2} = \sqrt{2500 - 900} = \sqrt{1600} = 40$ ft from the center.

12 Why can a whisper at one focus of an elliptical gallery be heard clearly at the other focus?

a. **Sound waves reflecting off the elliptical wall all converge simultaneously at the other focus** ✓

b. Sound waves travel faster along ellipses

c. The walls absorb echo

d. The ceiling acts as a resonator

By the reflective property of the ellipse, any ray emitted from one focus reflects off the perimeter directly toward the opposite focus with equal path length.

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